

EMMamba: An Efficient Medical Segmentation Model With Edge Enhancement

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1. Introduction: Context and Motivation

Medical image segmentation is a critical process for disease diagnosis, treatment planning, and prognosis assessment, as it allows physicians to precisely identify and localize anatomical structures and lesions. However, the inherent complexity of medical images—characterized by low contrast, high noise levels, variations in patient anatomy and imaging protocols, and the frequent occurrence of fuzzy or indistinct boundaries, especially in pathological cases—poses a significant challenge to achieving accurate and reliable segmentation. Traditional approaches, while foundational, have struggled to capture both detailed local features and the necessary global context efficiently. This has led to a search for architectures that can overcome these specific limitations.

- **The Core Innovation:** EMMamba is an efficient Mamba-based segmentation model for 2D medical images, leveraging Mamba's linear complexity for global context.
- **Edge & Boundary Focus:** It uniquely integrates an Edge-Aware Module (EAM) and a Boundary Sensitive Decoder (BSD) to handle fuzzy boundaries and low contrast issues, which are the main failure points for other models in tasks like polyp segmentation.
- **Superior Results:** It achieves state-of-the-art performance on 8 medical image segmentation datasets across 4 different modalities (polyp, breast, lung, brain). On the most challenging polyp dataset (ETIS), it achieves a Dice score of 72.22, outperforming all previous models.

2. Methodology: The EMMamba Architecture

The proposed EMMamba model is a novel multi-scale feature extraction framework built upon the Mamba backbone network. Its

innovation lies in the integration of three core modules specifically designed to address the challenges of medical images, particularly long-range dependency modeling and precise boundary delineation.

2.1. Overview: Efficiency of State Space Models

The model is inspired by the efficiency of Mamba's State Space Modeling (SSM). Unlike Convolutional Neural Networks (CNNs), which are efficient at capturing local spatial features but struggle with long-range dependencies, or Vision Transformers (ViTs), which excel at global context capture but are computationally heavy due to quadratic complexity, SSMs offer a promising alternative. They are designed to model long-range interactions with linear computational complexity, making them much faster and more efficient than Transformers. The Architecture of *EMMamba* is organized into four main components.

2.2. The Mamba Backbone Encoder

At its core, *EMMamba* utilizes a Mamba backbone network. This encoder processes the input medical image and extracts feature representations at multiple scales.

2.3. Core Module 1: The Efficient Fusion Bridge (EFB)

The EFB module is designed to handle the challenge of multi-scale feature integration. In many segmentation models, skip connections merely copy features from the encoder to the decoder, which can lead to a loss of information and sub-optimal fusion.

The EFB employs a more sophisticated mechanism:

- **Bidirectional State-Space Model (Bi-SSM):** It uses a Bi-SSM and attention mechanisms to intelligently fuse multi-scale features from different levels of the encoder. This allows the model to selectively combine high-level semantic information (like the shape of an organ) with low-level spatial details (like textures and edges).
- **Feature Harmonization:** This helps in harmonizing the features from different scales, ensuring that the decoder receives rich, complementary information for more accurate mask reconstruction.

2.4. Core Module 2: The Edge-Aware Module (EAM)

A key innovation is the Edge-Aware Module, which directly addresses the problem of fuzzy or indistinct boundaries. EAM enhances low-level edge representation by using Sobel-based edge extraction. The Sobel operator is a classic image processing technique used for edge detection. By integrating this into the deep learning pipeline, the model is explicitly guided to pay more attention to the structural outlines of objects. This forces the network to track the shift in tissue structure rather than relying solely on color contrast, which is particularly beneficial for polyp segmentation where the color of the polyp often blends with the surrounding tissue.

2.5. Core Module 3: The Boundary Sensitive Decoder (BSD)

Complementing the EAM, the Boundary Sensitive Decoder is responsible for generating the final, precise segmentation mask. It leverages a technique called "inverse attention" combined with residual convolutional layers to handle cross-level complex boundaries. Instead of focusing on the easy, often less informative center of an object, this module learns to force the network's processing power to concentrate on the more challenging boundary pixels. This ensures that the final mask is not only accurate in its core but also follows the exact contours of the object with high fidelity.

3. Experimental Setup and Results

To validate its performance, the model was put through a rigorous and thorough evaluation process.

3.1. Datasets

The authors evaluated their model on a total of 8 diverse medical segmentation datasets to demonstrate its wide applicability. These included:

- Polyp Segmentation: Kvasir, CVC-ClinicDB, CVC-ColonDB, CVC-EndoScene, and ETIS (the most challenging one).
- Breast Ultrasound Segmentation: BUSI dataset.
- Lung Lesion Segmentation: COVID-19 Lung dataset.
- Brain MRI Segmentation: LGG (Low-Grade Glioma) dataset.

3.2. Key Results

The ~~Æ~~Mamba model consistently achieves state-of-the-art (SOTA) performance across all tested medical imaging segmentation tasks. The following two tables summarize its

performance for the polyp and other medical segmentation tasks.

Table 1: Polyp Segmentation Results (mDice Score)

Dataset	Difficulty	ÆMMamba Score	Compared to
Kvasir-SEG	Standard	Highest mDice	Top baseline methods
CVC-ClinicDB	Standard	Highest mDice	Top baseline methods
CVC-ColonDB	Standard	Highest mDice	Top baseline methods
CVC-EndoScene	Standard	Highest mDice	Top baseline methods
ETIS	Most Challenging	72.22	Significantly outperforms MADGNet, Swin-UMamba

Table 2: General Medical Segmentation Results (Dice Score)

Task	Dataset	ÆMMamba Score	Compared to
Breast US Segmentation	BUSI	84.24	Surpasses H2Former, SwinUnet
Lung Lesion Segmentation	COVID-19 Lung	79.83	Surpasses H2Former, SwinUnet
Brain MRI Segmentation	LGG	87.25 / 79.31 mIoU	Outperforms all compared methods

4. Comparison with Competing Models

To highlight the leap in performance, it's helpful to understand what the new model is compared to. CNNs are fast but struggle with long-range dependencies; ViTs excel at global context but are computationally heavy. The table below provides a clearer side-by-side comparison.

Feature	CNNs (e.g., U-Net)	ViTs (e.g., SwinUnet)	ÆMMamba (Proposed)
Local Features	Excellent	Moderate	Excellent (via CNN modules)
Global Context	Poor	Excellent	Excellent (via SSM)
Boundary Precision	Moderate	Good	Excellent (via EAM & BSD)

Computational Efficiency	High ($O(n)$)	Low ($O(n^2)$)	High ($O(n)$)
Medical Edge Cases	Often fails on fuzzy boundaries	Struggles on low-contrast edges	Specifically designed to handle them

5. Why It Is SOTA for Polyp Segmentation

Polyps are widely considered one of the most difficult objects for AI to segment. `ÆMMamba` achieves SOTA status (including a massive 72.22 Dice score on the notoriously difficult ETIS dataset) because its architecture directly neutralizes the three main clinical challenges of a colonoscopy.

Challenge 1: The "Camouflage" Effect

- **The Problem:** Polyps are made of the exact same mucosal tissue as the colon wall. They are pink bumps on a pink background, meaning they lack high-contrast borders. CNNs constantly fail here, creating jagged or guessed boundaries.
- **The `ÆMMamba` Solution:** Because the EAM mathematically calculates gradients (edges) and the BSD uses inverse attention to focus on those edges, the model does not rely on color contrast to find the polyp. It tracks the structural shift of the tissue, allowing it to trace near-invisible borders flawlessly.

Challenge 2: Extreme Scale Variation

- **The Problem:** A colonoscope is a moving camera. A polyp might be a tiny 10-pixel speck in the background in one frame, and take up 90% of the screen in the next. Standard CNNs fail at this because their local "window" of computation can't understand the broader context of the image.
- **The `ÆMMamba` Solution:** The Vision Mamba (Vim) Encoder performs a selective scan across the whole image, giving it an infinite global view. It can correlate a tiny speck to the rest of the colon wall instantly, allowing it to segment polyps regardless of their size or distance from the camera.

Challenge 3: The Need for Real-Time Speed

- **The Problem:** For a segmentation model to actually be useful to a gastroenterologist, it must run live during the procedure at 30+ frames per second. Vision Transformers (ViTs) can achieve the accuracy needed, but their quadratic complexity makes them too slow and heavy for real-time video without massive supercomputers.
- **The ÆMMamba Solution:** Mamba State Space Models operate with linear computational complexity. It achieves the global accuracy of a heavy Transformer but processes math at the lightweight speed of a traditional CNN, making it perfectly suited for live clinical deployment.

6. Conclusion and Significance

ÆMMamba presents a compelling new paradigm for medical image segmentation by successfully integrating the computational efficiency of State Space Models with task-specific modules for edge and boundary enhancement. This design directly tackles the three main challenges of polyp segmentation—camouflage (solved by EAM's edge gradients and BSD's inverse attention), extreme scale variation (solved by the global view of the Vision Mamba encoder), and the need for real-time speed (solved by linear complexity). Its ability to process long-range dependencies with linear complexity, combined with its explicit focus on challenging boundaries, makes it a highly practical and powerful tool for real-time clinical applications. The paper's results clearly demonstrate its state-of-the-art status across a wide range of modalities and tasks, from low-contrast polyps to brain tumors.